Comparative Study of Matrix Riccati Equation Solvers for Parametric Faults Accommodation

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Abstract— **Linear quadratic fault accommodation is addressed. Under process and/or actuator faults, fault accommodation procedures need the controller to be redesigned, which amounts to solving the matrix algebraic Riccati equation (ARE) associated with the post-fault system model. In this respect, direct and iterative ARE solvers are analyzed, providing two different ways of application, namely the progressive respectively classical accommodation, function whether the ARE solution is used either during or after convergence. The longitudinal control of an aircraft under actuator and structural faults is used to illustrate the approaches.**

I. Introduction

Fault Tolerant Control (FTC) has established itself as an important research area in the last decade. Precisely, Fault Tolerance (FT) refers to the mechanism by which a given system is able to exhibit desired properties, both in normal operation and in the presence of faults occurring in sensors, actuators, or in the plant as well [1].

Properties of interest in the analysis of the nominal and post-fault systems range from stability to optimality, which can be expressed in terms of the minimum time and/or the minimum energy needed by the system transfer from one state to another [2]; for systems that can be described by a linear model, analyzing optimality can be done using the Linear Quadratic (LQ) control technique. Regarding the post-fault behavior, an automatic control system is said to be linear quadratic fault tolerant as it provides that a quadratic objective can still be satisfied in the presence of faults belonging to a given set.

Obviously, in both cases, respectively nominal and faulty, the optimal control law of a system is computed upon the solution of the corresponding matrix algebraic Riccati equation (ARE) [3]; at any rate, the direct and iterative solvers to the matrix ARE equally show interesting properties. However, in order to reduce the impairment of the post-fault system, the successive application of ARE solutions provided at each step by an iterative technique is recommended and expected to improve the closed-loop behavior.

In another paper [4], progressive accommodation was introduced by using the Newton-Raphson iterative technique to solve the ARE; however, other techniques might be used.

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This paper interrogates the classical and progressive fault accommodation abilities of several matrix ARE solvers, concluding that the progressive accommodation beats the classical scheme w.r.t. the closed-loop overall behavior; moreover, among the progressive approaches, it is shown that the Newton-Kleinman (NK) method, as it possesses the anytime property, is better than the Matrix-Sign-Function (MSF) technique; an application to the longitudinal control of an aircraft is developed as well.

It is organized as follows: Section II introduces the nominal and post-fault linear quadratic control problems. Further, Section III presents the direct and iterative methods to solving the matrix algebraic Riccati equation. The classical and progressive accommodation schemes are sketched in Section IV, while Section V states the numerical results and shows the simulations. Section VI concludes the paper.

II. Nominal and Post-Fault Linear Quadratic Control

A. Nominal Operation

Consider the continuous linear time-invariant (LTI) deterministic system whose nominal operation is modeled by

$$M_n : \dot{x}(t) = A_n x(t) + B_n u_n(t)\,, \text{ with } x(t_0 = 0) = x_0\,, \qquad (1)$$

where $A_n \in \mathbb{R}^{n\times n}$ and $B_n \in \mathbb{R}^{n\times m}$ are the nominal system and control matrices, $x(t) \in \mathbb{R}^n$ and $u(t) \in \mathbb{R}^m$ are the state and control vectors, and $x_0 \in \mathbb{R}^n$ is the initial condition.

Define the quadratic cost functional as

$$J(x_0, u(t)) = \int_0^\infty \left[x^T(t)Qx(t) + u^T(t)Ru(t)\right] dt\,, \qquad (2)$$

where $R > 0$ and $Q = \bar{Q}^T\bar{Q} \geq 0$ are weighting symmetric matrices, and $(.)^T$ stands for transposition.

Under the assumptions of stabilizability and detectability of pairs (A_n, B_n) and $(A_n, \bar{Q})$ respectively, the nominal linear state-feedback control law

$$u_n(t) = -K_n x(t)\,, \text{ with } K_n = R^{-1}B_n^T X_n\,, \qquad (3)$$

is optimal, where $X_n \in \mathbb{R}^{n\times n}$ is the unique symmetric semipositive-definite solution of the nominal continuous algebraic Riccati equation (CARE)

$$CARE_n : A_n^T X_n + X_n A_n - X_n B_n R^{-1} B_n^T X_n + Q = 0\,. \qquad (4)$$

In the nominal situation, the closed-loop system

$$\hat{M}_n : \dot{x}(t) = \hat{A}_n x(t)\,, \text{ with } \hat{A}_n = A_n - B_n K_n\,, \qquad (5)$$

is stable; provided the solution X_n to (4), the quadratic cost in (2) takes the minimum value

$$J_{M_n}(x_0, u_n) \triangleq J_n = x_0^T X_n x_0 \; (\equiv J^\star)\,. \qquad (6)$$

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B. Post-Fault Operation

Parametric faults being considered, the system to be controlled is described by the pair $(A_f, B_f) \neq (A_n, B_n)$, assuming that an LTI model is still able to capture its post-fault behavior and an online Fault Detection, Identification and Estimation (FDIE) module provides (A_f, B_f).

In this respect, the best strategy consists of applying the optimal control solution to the system

$$M_f : \dot{x}(t) = A_f x(t) + B_f u_f(t), \text{ with } x_{t_f} = x(t_f) \triangleq x_f, \quad (7)$$

on the post-fault time interval, i.e., $t \in [t_f, \infty)$.

Finding the optimal control associated with the faulty system is a controller redesign problem; it therefore amounts to solving the matrix CARE associated with the new situation, i.e.,

$$CARE_f : A_f^T X_f + X_f A_f - X_f B_f R^{-1} B_f^T X_f + Q = 0. \quad (8)$$

Assuming that the pairs (A_f, B_f) and $(A_f, \bar{Q})$ are still stabilizable and detectable respectively, the accommodated post-fault control law calculates as

$$u_f(t) = -K_f x(t), \text{ with } K_f = R^{-1} B_f^T X_f, \quad (9)$$

and, similarly to (5), the post-fault stable closed-loop system behavior is modeled by

$$\hat{M}_f : \dot{x}(t) = \hat{A}_f x(t), \text{ with } \hat{A}_f = A_f - B_f K_f; \quad (10)$$

furthermore, the control law (9) leads the cost functional (2) to the value

$$J_{M_f}(x_f, u_f) \triangleq J_f = x_0^T X_n x_0 + x_f^T (X_f - X_n) x_f. \quad (11)$$

C. Post-Fault Accommodation

Regarding the post-fault behavior (7)-(11), the primary interest resides in the analysis of methods to solving the matrix continuous algebraic Riccati equation associated with the faulty operation.

In this respect, two kinds of techniques are described, namely direct and iterative, precisely the Schur-Vector (SV) and Eigenvector (EV) methods for the direct approach, respectively the Newton-Kleinman (NK) and Matrix-Sign-Function (MSF) for the iterative one.

III. Direct and Iterative Matrix Riccati Equation Solvers

A. Direct Methods for Fault Accommodation

1) Schur-Vector Method: Recall that applying the QR-algorithm on the Hamiltonian-matrix

$$H_f = \begin{bmatrix} A_f & -G_f \\ -Q & -A_f^T \end{bmatrix}, \text{ with } G_f = B_f R^{-1} B_f^T, \quad (12)$$

drives it to the upper (quasi-)triangular Real Schur Form (RSF) (see, e.g., [5]); denote it by $\tilde{H}_f = U_f^T H_f U_f$, where $H_f \in \mathbb{R}^{2n \times 2n}$; if H_f has no eigenvalues on the imaginary-axis, a stabilizing solution to $CARE_f$ still exists (see, e.g., [6]).

Partitioning $\tilde{H}_f$ and the matrix of accumulated transformations U_f conformably with H_f, the first n columns of U_f, precisely $[\ U_{f_{11}}^T \quad U_{f_{21}}^T\ ]^T$, span the stable invariant-subspace of the Hamiltonian-matrix (12).

In this case, the stabilizing solution to $CARE_f$ computes as

$$X_f = U_{f_{21}} U_{f_{11}}^{-1}$$

and $\Lambda(\tilde{H}_{f_{11}}) = \Lambda(A_f - G_f X_f) \subset \mathbb{C}^-$ (see [7]), where $\Lambda(.)$ stands for the spectrum of the matrix-argument.

Assuming that the average QR-iterations per eigenvalue is 2 and including the RSF ordering of the Hamiltonian-matrix, the work-count for the Schur-Vector method is nearly estimated at $200n^3$ flops (see, e.g.,[8]).

2) Eigenvector Method: Alternatively, a similar expression for X_f may be established by using the Eigenvector method; write it generically as $X_f = Y_{f_2} Y_{f_1}^{-1}$, where Y_{f_1} and Y_{f_2} are real $n \times n$ matrices formed by the upper and lower halves of eigenvectors or generalized-eigenvectors in a Jordan-chain of H_f chosen such that Y_{f_1} is invertible [9].

Though less computationally expensive (the flop-count for this approach stands for the lower ballpark $160n^3$) than the SV method (see again [8]), it appears that the Eigenvector method may become unstable for multiple eigenvalues and the Schur-Vector method is better numerically conditioned.

The major drawback of direct methods is that, during the computation time of the solution to $CARE_f$, the closed-loop stability and performance are uncertain since the post-fault system is under the nominal control. Therefore, it is worth investigating iterative techniques to solving the post-fault CARE (8), in order to improve the global closed-loop behavior and possibly shorten the control accommodation period.

B. Iterative Methods for Fault Accommodation

1) Matrix-Sign-Function Approach: Reconsider the Hamiltonian-matrix (12) and note that X_f is the desired solution to the post-fault CARE iff

$$\begin{bmatrix} A_f & -G_f \\ -Q & -A_f^T \end{bmatrix} = \begin{bmatrix} I_n & 0_n \\ X_f & I_n \end{bmatrix} \begin{bmatrix} \hat{A}_f & -G_f \\ 0_n & -\hat{A}_f^T \end{bmatrix} \begin{bmatrix} I_n & 0_n \\ X_f & I_n \end{bmatrix}^{-1},$$

where $\hat{A}_f = A_f - B_f K_f = A_f - G_f X_f$ is the closed-loop matrix in the faulty case, with I_n and 0_n respectively the identity and null matrices.

Henceforth, introduce the matrix

$$J = \begin{bmatrix} 0_n & I_n \\ -I_n & 0_n \end{bmatrix} \in \mathbb{R}^{2n \times 2n}$$

and observe that H_f is J-symmetric, i.e., $JH_f = (JH_f)^T$; that is if $\lambda_f \in \Lambda(H_f)$ then $-\lambda_f \in \Lambda(H_f)$. Hence, the spectrum of the Hamiltonian-matrix is symmetric w.r.t. the imaginary-axis and therefore H_f has exactly n stable eigenvalues.

Remember also that there always exists a similarity transformation T_f of eigenvectors such that

$$H_f = T_f \begin{bmatrix} J_f^- & 0_n \\ 0_n & J_f^+ \end{bmatrix} T_f^{-1}, \quad (13)$$

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where J_f^- and J_f^+ are Jordan-blocks associated with the stable and unstable eigenvalues respectively.

The sign-function matrix of H_f may then be computed upon (13) as follows (see, e.g., [10])

$$\operatorname{sign}(H_f) = T_f \begin{bmatrix} -I_n & 0_n \\ 0_n & I_n \end{bmatrix} T_f^{-1}; \tag{14}$$

note that $\Lambda(\operatorname{sign}(H_f))$ contains only ± 1's, which means the sign-function matrix (14) is involutory, i.e., $(\operatorname{sign}(H_f))^2 = I_{2n}$.

Thus, the Newton iteration for computing the principal square-root of the identity matrix

$$Z_{f_{k+1}} = \frac{1}{2}\left(Z_{f_k} + Z_{f_k}^{-1}\right), \text{ for } k = 1,2,\ldots,N, \tag{15}$$

with the initial step $Z_{f_0} = H_f$, obeys $\lim_{k\to\infty} Z_{f_k} = \operatorname{sign}(H_f) = Z_{f_N}$ (see, e.g., [11]).

Moreover, accelerating the speed of the first steps is possible through an appropriate scaling [12] and increasing efficiency is reached by symmetrizing the iterations [13], that is replacing (15) with

$$\bar{Z}_{f_k} = Z_{f_k}\left(\det(Z_{f_k})\right)^{-\frac{1}{2n}}, \tag{16a}$$

$$Z_{f_{k+1}} = \bar{Z}_{f_k} - \frac{1}{2}\left(\bar{Z}_{f_k} - (J\bar{Z}_{f_k})^{-1}J\right). \tag{16b}$$

Partitioning Z_{f_k} accordingly, the solution X_{f_k} to (8) results then by solving the system

$$\begin{bmatrix} Z_{f_k}^{12} \\ Z_{f_k}^{22} + I_n \end{bmatrix} X_{f_k} = -\begin{bmatrix} Z_{f_k}^{11} + I_n \\ Z_{f_k}^{21} \end{bmatrix}, \tag{17}$$

for $k = N$ (see, e.g., [14]).

The computational-load required by the Matrix-Sign-Function to obtaining the solution to the post-fault CARE is averaged at $4n^3$ per iteration (see [8]). (Anyhow, the convergence-speed of the MSF technique depends mostly on the eigenvalues magnitudes and not on the system order n; see [15].)

However, the complexity of this approach might be considerably raised whether solving at each step the over-determined system of equations (17), solution of which is better numerically conditioned if obtained using the QR-algorithm instead of the pseudo-inverse matrix.

2) Newton-Kleinman Approach: Let X_{f_k} be the unique semipositive-definite solution of the algebraic Lyapunov equation

$$\hat{A}_{f_k}^T X_{f_k} + X_{f_k}\hat{A}_{f_k} + K_{f_k}^T R K_{f_k} + Q = 0, \tag{18}$$

where, for $k = 1,2,\ldots,N$, recursively,

$$K_{f_k} = R^{-1}B_f^T X_{f_{k-1}},$$

with $\hat{A}_{f_k} = A_f - B_f K_{f_k}$ and K_{f_0} chosen such that the matrix $\hat{A}_{f_0} = A_f - B_f K_{f_0}$ has stable eigenvalues.

Then $\lim_{k\to\infty} X_{f_k} = X_{f_N}$, with the ordering $0 \le X_{f_N} \le \ldots \le X_{f_{k+1}} \le X_{f_k} \le \ldots \le X_{f_1}$ (see [16]). In general, $X_{f_0} \ge X_{f_1}$ is not true.

It follows that, given any initial stabilizing matrix K_{f_0}, for N large enough, the solution to the continuous algebraic Lyapunov equation

$$\hat{A}_{f_N}^T X_{f_N} + X_{f_N}\hat{A}_{f_N} + K_{f_N}^T R K_{f_N} + Q = 0 \tag{19}$$

converges to that of the continuous algebraic Riccati equation

$$A_f^T X_{f_N} + X_{f_N} A_f - K_{f_N}^T R K_{f_N} + Q = 0; \tag{20}$$

removing index N, remark that (20) and (8) are identical.

The computational-effort accumulated by 8-10 iterations performed by the Newton-Kleinman method, usually sufficient to solving the post-fault CARE, is equivalent to that provided by the Schur-Vector method directly used in the same scope.

C. Initial Step for Newton-Kleinman Method

If the system remains stable after a fault, the iterative accommodation procedure to the new situation obviously can be initialized using the solution to $CARE_n$, i.e., using X_n which gave the nominal control matrix K_n until the moment the fault occurred.

Nonetheless, in situations when the closed-loop matrix $\hat{A}_f = A_f - B_f K_n$ becomes unstable after the fault, an appropriate stabilizing initialization is crucial in order to ensure convergence.

1) Kleinman Initialization Method: If the pair (A_f, B_f) remains completely controllable, then $u_f(t) = -K_f x(t)$ is a stabilizing control law with K_f given, for arbitrary T, by (see [17])

$$K_f = B_f^T W_f^{-1}(T), \tag{21a}$$

$$\text{with } W_f(T) = \int_0^T e^{-A_f t} B_f B_f^T e^{-A_f^T t}\, dt. \tag{21b}$$

Unfortunately, choosing the value of T in (21b) is a question of fault criticality; it requires a tradeoff to be performed between the accuracy of the integration result and the time available for obtaining the initial stabilizing feedback.

Note however that $W_f(T)$ in (21b) represents the controllability Gramian of the system pair (A_f, B_f), i.e., the solution to the continuous algebraic Lyapunov equation

$$(-A_f)W_f + W_f(-A_f)^T + B_f B_f^T = 0 \tag{22}$$

when taking the integration limit $T \to \infty$, that is (22) provides the structure of the initial control matrix.

2) Bass-Armstrong Initialization Method: An extension to (22) firstly operates a shift on the diagonal elements of the post-fault system matrix A_f; in this case, $W_f = W_f^T > 0$ is found to satisfy

$$[-(A_f + \beta_f I_n)]W_f + W_f[-(A_f + \beta_f I_n)]^T + 2B_f B_f^T = 0, \tag{23}$$

with $\beta_f > \|A_f\|$, where $\|.\|$ defines any matrix-norm subordinated to a consistent vector-norm; then, $K_f = B_f^T W_f^{-1}$ stabilizes the pair (A_f, B_f) (the choice $\beta_f > \|A_f\|$ counteracts the instability of A_f, ensuring that matrices $(A_f + \beta_f I_n)$ and $-(A_f + \beta_f I_n)$ share no common eigenvalues and thus (23) has a solution) [18].

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Still, despite the restrictive assumption of complete controllability of (A_f, B_f), similar results relax this condition to stabilizability only [19]; in that case, W_f^{-1} is replaced by the pseudo-inverse matrix $W_f^{\dagger}$.

3) Bartels-Stewart Matrix Lyapunov Equation Solver: At the core, algorithms to obtaining the solution to the initial step problem and the intermediate solutions of the Newton-Kleinman approach basically demand for rapid and numerically stable solvers of matrix Lyapunov equations; in this respect, the Bartels-Stewart algorithm is considered the most efficient one [20].

The total amount of work performed by the Bartels-Stewart algorithm is averaged at $14n^3$.

IV. Classical vs. Progressive Fault Accommodation

After all, several different ways of accommodating the post-fault system are foreseen, that is using either the solution provided by the direct matrix CARE solvers respectively the one obtained after the iterative approaches achieved convergence or, alternatively, using the solution provided at each step by the iterative techniques.

Hence, it calls for two kinds of post-fault accommodation schemes, namely classical and progressive, both being sketched in what follows.

A. Classical Accommodation Scheme

Assume that N is the number of steps needed to obtain a satisfactory approximation either for the solution to (16) or (19) and take Δ_k the duration of iteration k.

A classical accommodation procedure would naturally apply the nominal control law (3) during the period $t_{ftc} - t_{fdie} = \bar{\Delta}$ $\left(\bar{\Delta} = \sum_{k=1}^{N} \Delta_k \text{ and } \bar{\Delta}_k = \sum_{l=1}^{k} \Delta_l\right)$ until the accommodated feedback is available, where t_{fdie} is the time at which the new post-fault pair (A_f, B_f) is given by the FDIE module while t_{ftc} is the instant at which the new post-fault control matrix K_f computed by the FAR (Fault Accommodation/Reconfiguration) module becomes available. (Assimilate $\bar{\Delta}$ also with the computation time needed by the SV method.)

Hence, using the Classical Accommodation Scheme results in a two-stage post-fault behavior, as sketched in Algorithm 1.

Algorithm 1: Classical Accommodation Scheme (CAS)

(1-1) While computing the accommodated control law,

$$\dot{x}(t) = (A_f - B_f K_n)x(t)\,, \text{ for } t \in [t_f, t_{fdie} + \bar{\Delta})\,.$$

(1-2) When iterations reached convergence,

$$\dot{x}(t) = (A_f - B_f K_{f_N})x(t)\,, \text{ for } t \in [t_{fdie} + \bar{\Delta}, \infty)\,.$$

B. Progressive Accommodation Scheme

Contrary to the classical scheme, the progressive accommodation approach would apply the control laws $u_{f_k}(t) = -K_{f_k}x(t)$ for $k = 1, 2, \ldots, N$, with K_{f_k}'s computed upon the solutions X_{f_k} provided either by (17) or (18), rather than waiting for the complete convergence of iterations (see also [4]).

This time, using the Progressive Accommodation Scheme results in a three-stage post-fault behavior, as sketched in Algorithm 2.

Algorithm 2: Progressive Accommodation Scheme (PAS)

(2-1) During the first iteration,

$$\dot{x}(t) = (A_f - B_f K_n)x(t)\,, \text{ for } t \in [t_f, t_{fdie} + \Delta_1)\,.$$

(2-2) While progressive accommodation proceeds,

$$\dot{x}(t) = (A_f - B_f K_{f_k})x(t)\,, \text{ for } t \in [t_{fdie} + \bar{\Delta}_k, t_{fdie} + \bar{\Delta}_{k+1})\,.$$

(2-3) When iterations reached convergence,

$$\dot{x}(t) = (A_f - B_f K_{f_N})x(t)\,, \text{ for } t \in [t_{fdie} + \bar{\Delta}, \infty)\,.$$

Remark 1: Note the resemblance of PAS with the Extended Linearization Control Method used to solving the State-Dependent Riccati Equation (SDRE) with State-Dependent Coefficient (SDC) matrices, frozen within the intervals of effective computation, in spite the variation of the state (see, e.g., [21]).

C. Anytime Accommodation Algorithms

Whether considering to apply efficiently one of the two iterative techniques in PAS, either Newton-Kleinman or Matrix-Sign-Function, it must be noted that the monotonic non-increasing sequence of solutions provided by NK conducts also to an equivalent monotonically decreasing series of the intermediate optimal costs, while the alternate-sign in the definiteness of solutions yielded by MSF does not provide a clear evolution of the cost spent for control accommodation.

As it improves the control solution at each step, the Newton-Kleinman method therefore possesses the anytime property and the Matrix-Sign-Function does not (see [22]; see also [23]).

In the sequel, the design of a linear quadratic controller for the longitudinal model of a civil aircraft is considered, with the aim of proving the advantages of using the progressive accommodation approach after a structural and an actuator fault. (See also [24].)

V. LQ Fault Accommodation of the B747 Aircraft Longitudinal Model

A. Flight Condition

Consider a Class III large transport aircraft Boeing 747 in Class I cruise flight. Let the Category B flight phase be, i.e., gradual maneuvering without precision tracking.

Assume a straight and constant level flight at 40000 ft fixed altitude and 0.8 Mach; the corresponding steady-state speed is 774 fps. (The CoG position is $(0.25, 0, 0)$ m.a.c. and the aircraft mass is 19792 $slug$.)

B. Nominal System

1) Open-Loop Model: The aircraft longitudinal dynamics is described by the state vector $x = \begin{bmatrix} u & w & q & \theta \end{bmatrix}^T$, where u (ft/sec) and w (ft/sec) represent the inertial velocities in the x- and z- directions of F_B reference frame;

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also, q (rad/sec) and θ (rad) represent respectively the pitch rate and the pitch angle (F_B denotes body-axis reference). The control input δ_E (rad) is the elevator deflection.

The linearized model of the system is given by the nominal pair of system and control matrices (A_n, B_n) as (see [25])

$$A_n = \begin{bmatrix} -0.0069 & 0.0139 & 0.0 & -32.2000 \\ -0.0904 & -0.3147 & 773.9766 & 0.0 \\ 0.0001 & -0.0010 & -0.4284 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \end{bmatrix},$$

$$B_n = \begin{bmatrix} -0.0002 & -18.0610 & -1.1577 & 0.0 \end{bmatrix}^T.$$

2) Closed-Loop System: To design the full-state linear quadratic regulator, the penalty matrices for controls and the states are chosen to be $Q = 10^{-8} \times I_4$ and $R = 10^{-4}$.

The nominal control matrix

$$K_n = \begin{bmatrix} 0.0076 & -0.0047 & -2.9529 & -3.0015 \end{bmatrix} \quad (24)$$

provides the closed-loop system $\hat{M}_n$, with $\hat{A}_n = A_n - B_n K_n$, which has the stable spectrum

$$\Lambda(\hat{A}_n) = \{-2.0578 \pm 2.2020\imath, -0.0687 \pm 0.0907\imath\}.$$

C. Faulty System

Consider an actuator fault defined by the loss-of-effectiveness with $\tau_f = 50\%$ in the elevator deflection; that can be modeled in the post-fault system by multiplying the control matrix with a factor $(1-\tau_f) = 0.5$, i.e., $B_f = (1-\tau_f)B_n$. Also, assume a structural fault modeled in the system matrix as $A_f = A_n + \Delta A_f \times I_4$, with $\Delta A_f = \sigma_n^{\max} \times 10^{-3}$, where $\sigma_n^{\max}$ is the maximum singular value of A_n.

D. Progressive Accommodation

1) Newton-Kleinman Approach: Note that, for the faulty pair (A_f, B_f) introduced before, the nominal control matrix K_n given in (24) does not ensure stability anymore.

The post-fault closed-loop system $\hat{M}_f$, with $\hat{A}_f = A_f - B_f K_n$, has got an unstable pair of poles in the spectrum

$$\Lambda(\hat{A}_f) = \{-0.4139 \pm 1.8885\imath, 0.7110 \pm 0.0903\imath\}.$$

Use therefore the Bass-Armstrong approach with a scaled $\beta_f = 0.7740$ and obtain the initial control matrix

$$K_{f_0} = \begin{bmatrix} 7.0007 & -0.3173 & -4.4520 & 203.9306 \end{bmatrix}$$

that provides the stable closed-loop system $\hat{M}_{f_0}$, with $\hat{A}_{f_0} = A_f - B_f K_{f_0}$, with two pairs of complex-conjugate poles

$$\Lambda(\hat{A}_{f_0}) = \{-0.7740 \pm 3.3699\imath, -0.7740 \pm 0.6729\imath\}.$$

To decide whether the progressive fault accommodation achieved convergence, a condition on the difference between the control gains in two successive iterations was used. The post-fault closed-loop needed $N = 8$ iterations for the accommodation. The evolution of the first control gain using Newton-Kleinman method is shown in Table I.

TABLE I
1ST CONTROL GAIN IN NEWTON-KLEINMAN APPROACH

k	1	2	3	4
$k_{f,1}^{nk,acc}$	7.007	12.2505	6.9017	4.4110
k	5	6	7	8
$k_{f,1}^{nk,acc}$	3.4114	3.1444	3.1205	3.1203

TABLE II
1ST CONTROL GAIN IN MATRIX-SIGN-FUNCTION APPROACH

k	1	2	3	4	5	6
$k_{f,1}^{msf,acc}$	0.0076	0.0100	2.0921	3.1280	3.0940	3.1203

2) Matrix-Sign-Function Approach: Recall that there is no need for a special initialization procedure in this case. Using the same criterion for testing convergence as previously, the method took only $N = 6$ steps for the accommodation of the post-fault closed-loop system. The evolution of the first control gain in the Matrix-Sign-Function approach is shown in Table II.

E. Accommodated System

1) Accommodated Closed-Loop: The progressive fault accommodation, either based on the Newton-Kleinman approach or the Matrix-Sign-Function method, converges to the control matrix

$$K_f^{acc} = \begin{bmatrix} 3.1203 & -0.0884 & -10.1884 & 33.0122 \end{bmatrix},$$

which is computed upon the solution to the accommodated post-fault CARE; then, the accommodated closed-loop system $\hat{M}_f^{acc}$, with $\hat{A}_f^{acc} = A_f - B_f K_f^{acc}$, has a stable accommodated spectrum

$$\Lambda(\hat{A}_f^{acc}) = \{-1.4041 \pm 1.6043\imath, -0.7707 \pm -0.0607\imath\}.$$

2) Closed-Loop Simulation: Fig. 1 displays the simulations of the closed-loop behavior during the post-fault period when applying the progressive and the classical control schemes. For being concise, the simulations reveal the accommodated behavior only of the inertial vertical velocity and that of the pitch rate respectively (short-period mode); the excitation is produced for 3 *sec* by a 5 *deg* magnitude doublet-pulse.

The fault occurred at $t_f = 4$ *sec*, the estimation of the faulty model was ready at $t_{fdie} = 5$ *sec*, therefore there is one sample of uncertain evolution, until $t_{fdie+\Delta} = 6$ *sec*, when the solution to the initial step problem will be readily available, this one being applied until $t_{fdie+2\Delta} = 7$ *sec* (for simplicity, the case of samples $\Delta_k \equiv \Delta = 1$ *sec* was considered).

Also, see in Fig. 1(b) the huge overshoot that produces when classical accommodation procedure is used. The closed-loop instability boils down during the convergence process such that the recovery graphs make the ones showing the progressive accommodation appear insignificant. (Compare the w- and q- scales on Figs. 1(a) respectively 1(b).)

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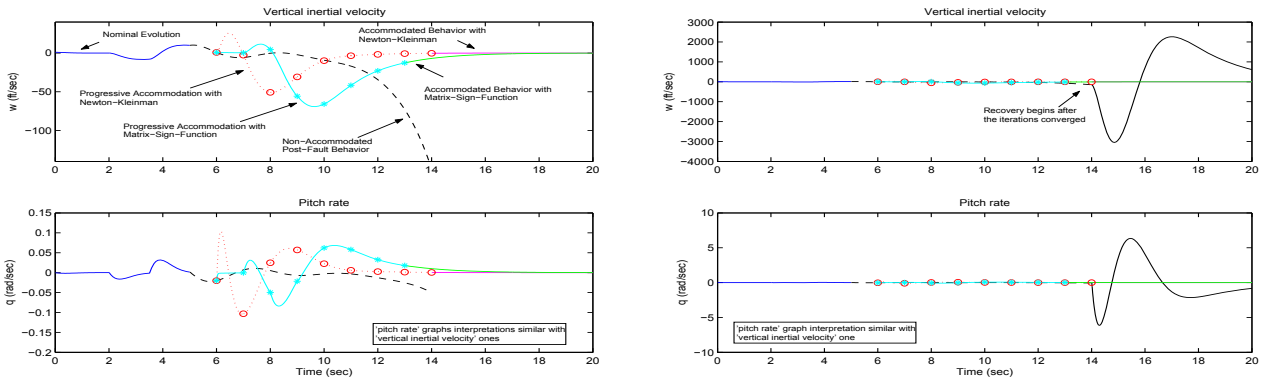

(a) Progressive Scheme (b) Classical Scheme

Fig. 1. Fault Accommodation of the B747 Short-Period Mode

TABLE III
COMPARATIVE STUDY ON MATRIX CARE SOLVERS

	SV	EV	NK	MSF
Classical	yes	yes	yes	yes
Progressive	no	no	yes	yes
Anytime	-	-	yes	no
Complexity	$\approx 200n^3$	$\approx 160n^3$	$\approx 14n^3/\text{it}$	$\approx (4n^3 + O(\text{QR}))/\text{it}$

VI. CONCLUSION

This paper interrogated several matrix continuous algebraic Riccati equation (CARE) solvers, namely the Schur-Vector (SV) and Eigenvector (EV) methods for the direct approaches, respectively the Newton-Kleinman (NK) and Matrix-Sign-Function (MSF) for the iterative ones, aiming at their integration into two types of techniques for fault accommodation, namely the classical and the progressive schemes.

It is concluded that the progressive accommodation beats the classical scheme as it improves the closed-loop overall behavior; moreover, among the progressive approaches, the NK method, as it possesses the anytime property, is better than the MSF technique. The synthesis of these methods in view of their application for fault accommodation is shown in Table III.

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